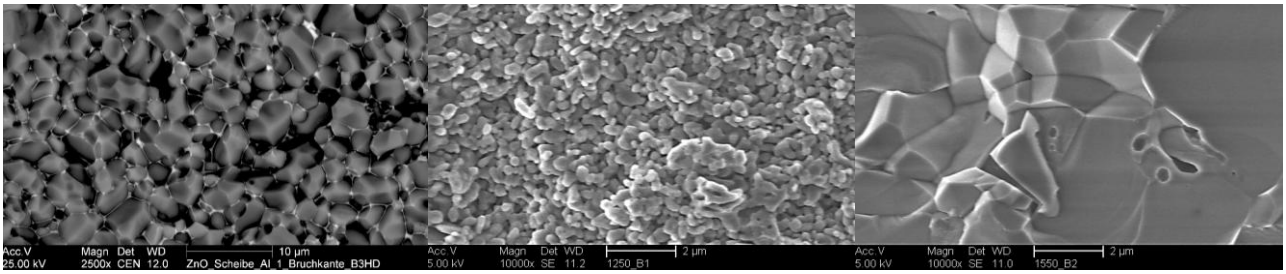
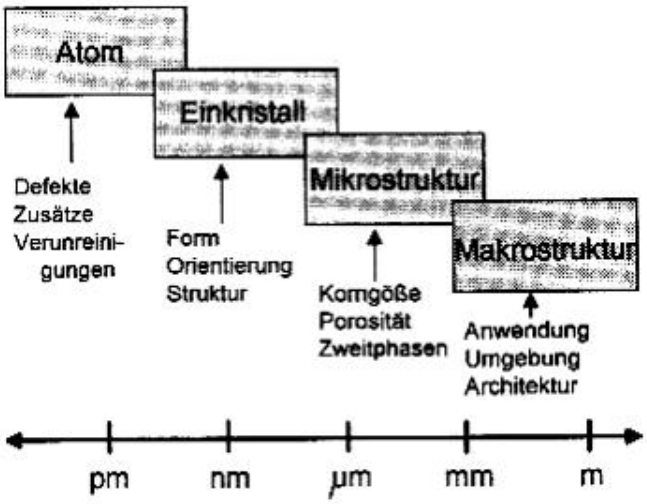
CERAMIC ENGINEERING

Chapter 10: selected ceramic materials



SEM images: Ulf Betke, OVGU, 2025

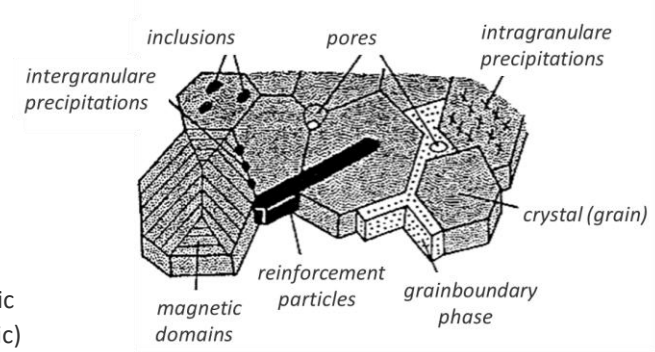
Curtesy of Chr. Aneziris and P. Gehre, Technical University of Freiberg, Germany



structural levels and interactions in ceramic materials

microstructure (general) of ceramics

- crystals
- interfaces (grain boundaries, phase boundaries, polarization boundaries)
- inclusions (dispersed particles, fibres)
- grain boundary phases (glassy phases, precipitations, segregations)
- flaws and faults (pores, cracks, inclusions)
- residual stresses



polycrystalline ceramic microstucture (schematic)

selected oxide ceramics

alumina	Al_2O_3
zirconia	ZrO_2
partially stabilized zirconia (w. CeO, CaO, MgO or Y_2O_3)	$Zr_{0.9}Mg_{0.1}O_{1.9}$
aluminum titanate	Al_2TiO_5 (ATi, AlTi)
cordierite	$Mg_2Al_4Si_5O_{18}$
mullite	$Al_6Si_2O_{11}$
spinel	$MgAl_2O_4$
lithium aluminum silicate	Li_2O-Al_2O_3-SiO_2 - Basis (LAS)

selected oxide ceramics

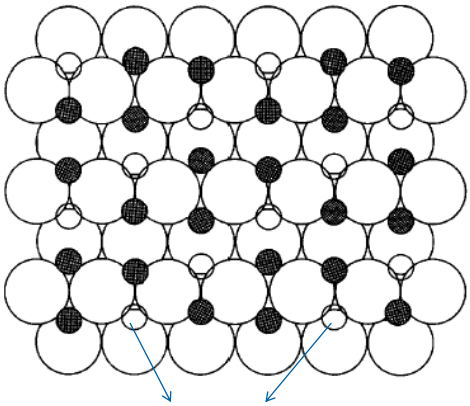
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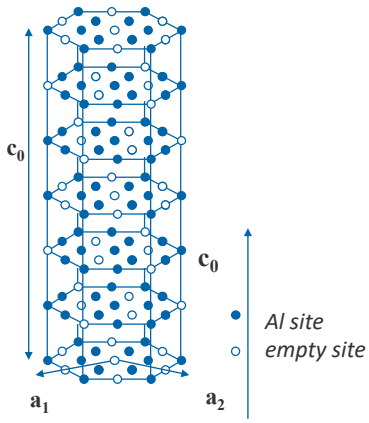
***alumina* (also: α-alumina, α-Al_2O_3, corundum, sapphire*)**

ρ $\left[\frac{g}{cm^3}\right]$	E [GPa]	K_{IC} $[MPa\sqrt{m}]$	σ_B [MPa]	m [1]
3.98	400	3.4	400	10

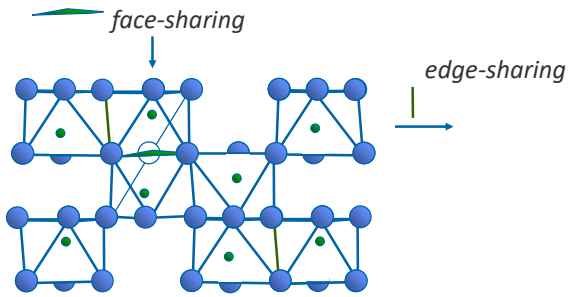
hardness [HV10]	α $[10^{-6}\ K^{-1}]$	λ $\left[\frac{W}{m \cdot K}\right]$	melting pt [°C]
2100	5.5-10	36 (RT)	2050

*pls. note: sapphire is the monocrystalline form of α-alumina; when sapphire is doped with chromium-III (Cr^{3+} ions) it appears deep red and is called ruby. Ruby is used in LASER applications





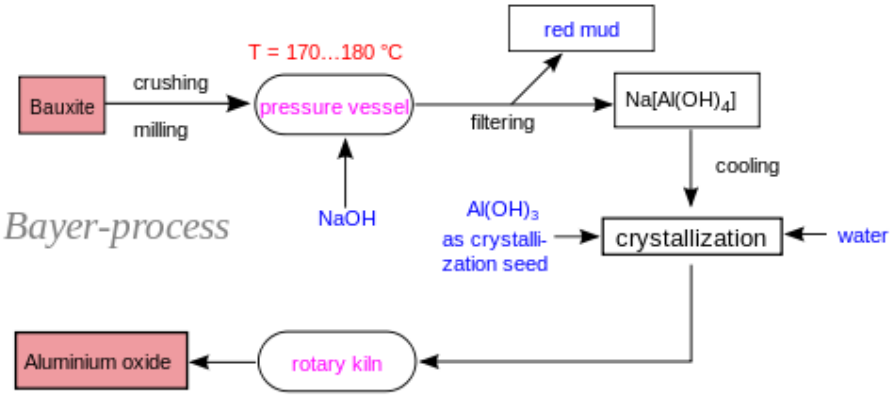
corundum structure, hexagonal unit cell setting;
only cationic sublattice shown.
oxygen forms a hexagonal dense packed array



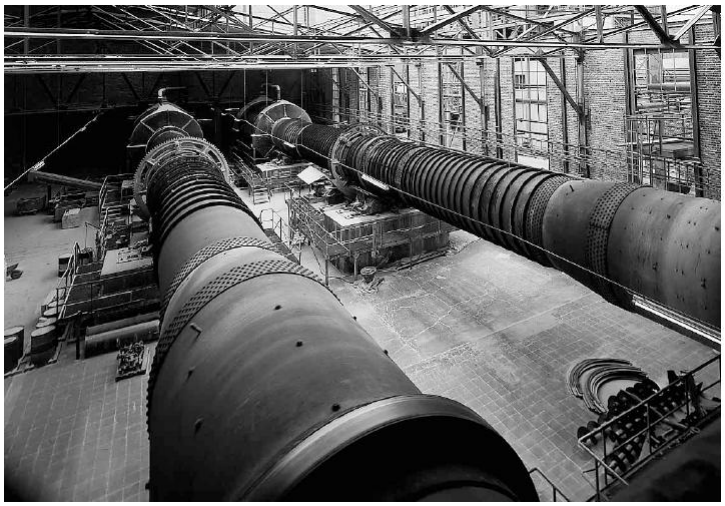
(210) projection of the corundum structure;
Al ions in adjacent face-sharing octahedra
mutually repell each other.

manufacturing of alumina

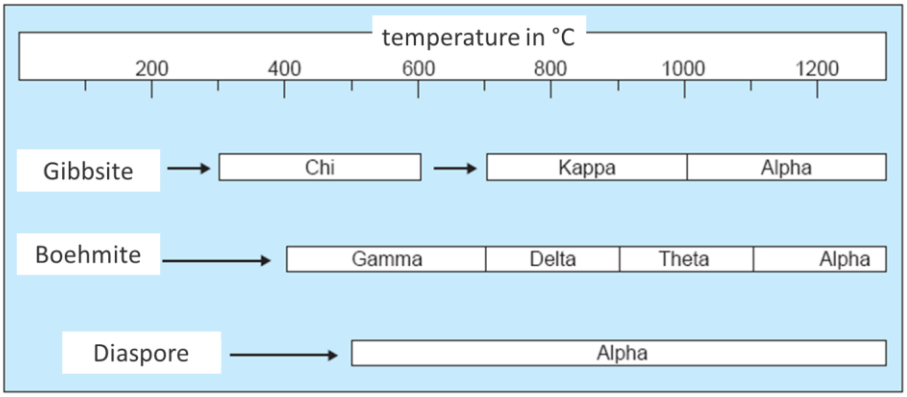
refining bauxite to
produce alumina
with the Bayer process

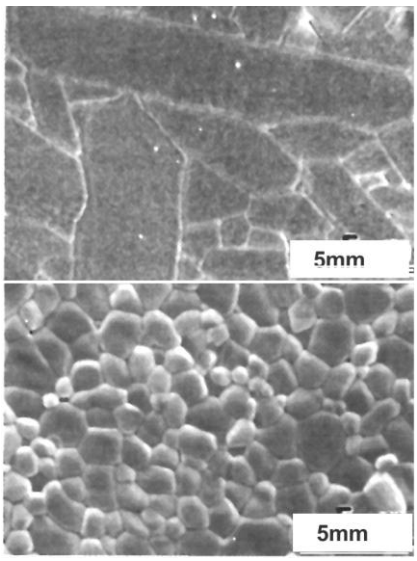


3,7 t bauxite ➔ 1,9 t alumina ➔ 1 t aluminum



types of hydrated aluminas prior to final phase change to α-alumina



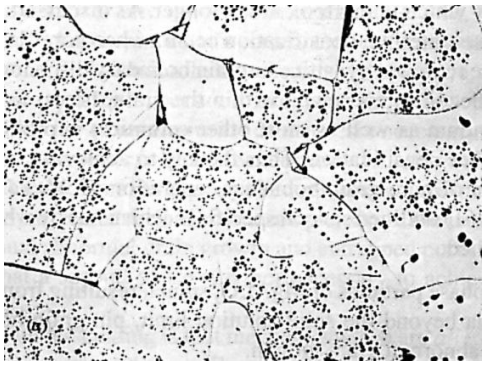


dense hot pressed alumina without (top) and with addition of MgO (bottom); grain growth is detrimental to the fracture strength of ceramics:

$$\sigma_f \propto d^{-1/2}$$

d: grain diameter

doping alumina with MgO leads to the formation of precipitates of spinell along the grain boundaries, which lowers the grain boundary mobility
(Bennison et al., 1983)



SEM image by Geskovich et al. shows an alumina body sintered without dopant

porosity is detrimental for the mechanical strength:

$$\sigma_f = \sigma_0 \exp^{(-bp)}$$

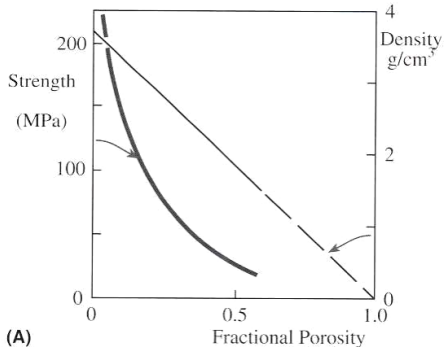
σ_0: strength at zero porosity
b: constant

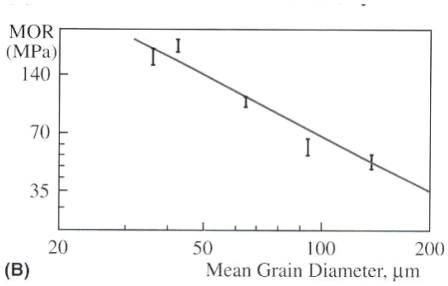
reduction of intragranular porosity:
doping with periclase (MgO)

- reduces the internal residual porosity: lower grain boundary mobility allows the filling of the pores when pores reach the grain boundaries
- fast grain growth encloses the pores quickly into the interior of the grain→difficult to eliminate

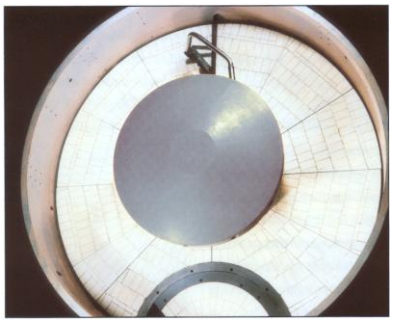
however, even if porosity is eliminated and grain size is small to moderate, a small Ki,c remains in the resulting alumina ceramic → reinforcement, e. g. by ZrO_2 doping **(ZTA)**

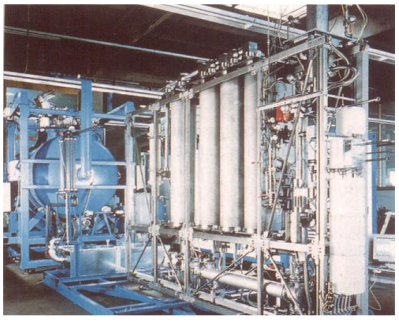
remember!





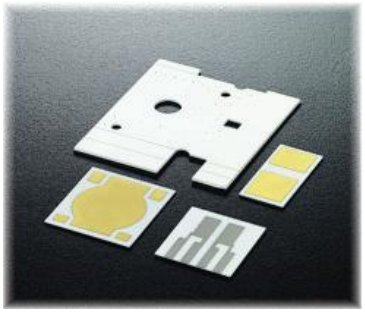







ETEC





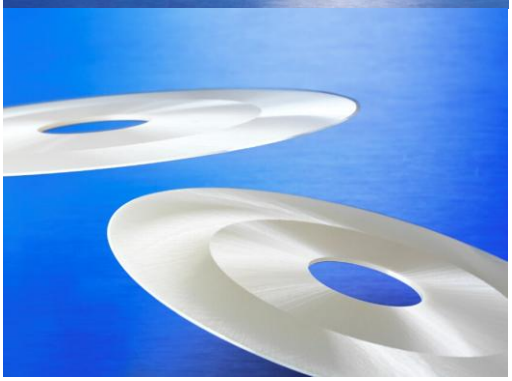




wearing parts
in textile industry

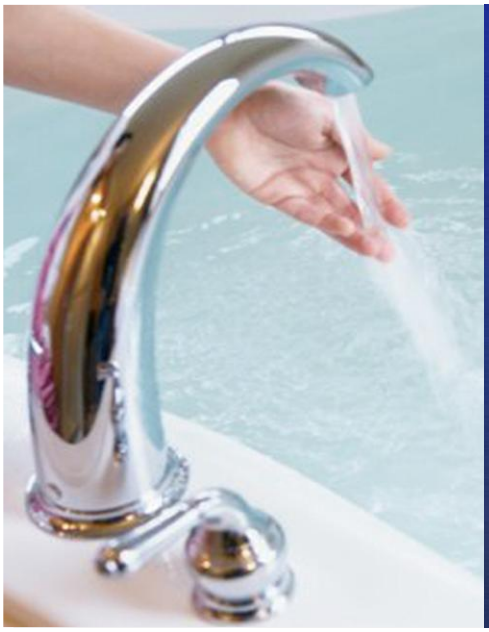


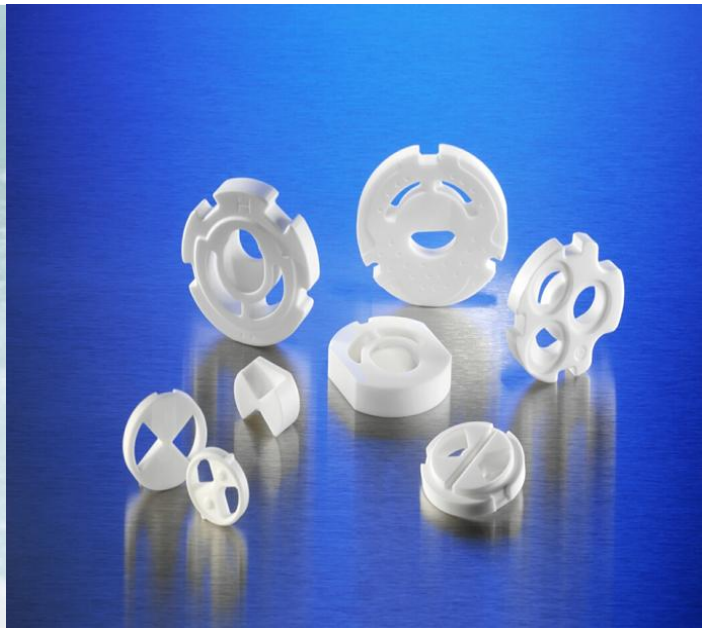
friction dsiks



Al_2O_3 cutters

CeramTec

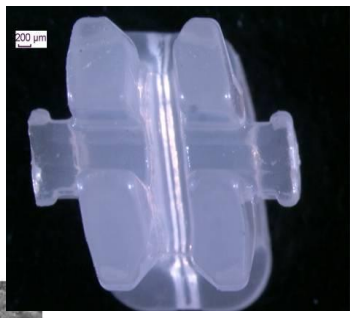




sealing and regulating washers

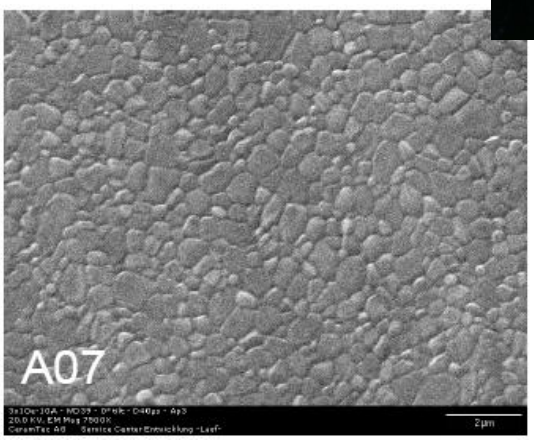
CeramTec

translucent alumina brackets



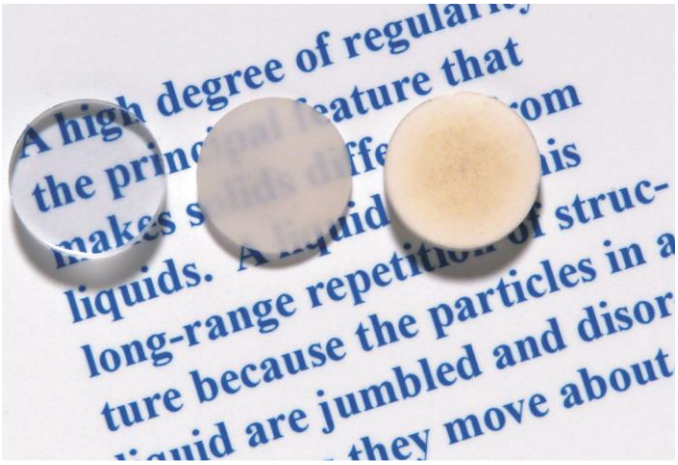
high pressure injection molding

density: 3.99 g/cm³
average grain size: 0.4 µm
Inline-transmission: 24 %
total transmission: 62 %
Flexural strength: 850 MPa
hardness: 2015 HV 10
Youngs modulus: 415 GPa
fracture toughness: 4.1 MPam$^{1/2}$



CeramTec

three thin disk specimens of aluminum oxide, placed over a printed page in order to demonstrate their differences in light-transmittance characteristics



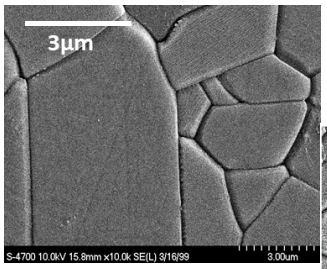
left: *transparent* (i.e., virtually all light that is reflected from the page passes through it),

center: *translucent* (meaning that some of this reflected light is transmitted through the disk).

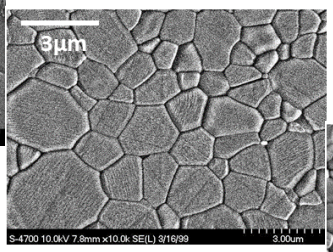
Right: *opaque;* none of the light passes through

differences in optical properties→a consequence of differences in microstructure of these materials, which have resulted from the way the materials were processed.

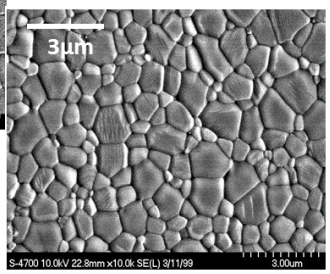
alumina: a microstructure´s brief history



Al_2O_3 from fine-particulate powder (before 1980)



alumina powders from the alcoxide process (since 1994)



powders from $NH_4AlCO_3(OH)_2$ decomposition (since 1999)

CeramTec

***zirconia* (also: zirkoniumoxid, ZrO_2)**

ρ $\left[\frac{g}{cm^3}\right]$	E [GPa]	K_{IC} $[MPa\sqrt{m}]$	σ_B [MPa]	m [1]
5.89	200	6-10	60-1000	15-25

Härte [HV10]	α $[10^{-6}\ K^{-1}]$	λ $\left[\frac{W}{m \cdot K}\right]$	Schmelzp. [°C]
1300	10	2	2680

crystallographic structures:

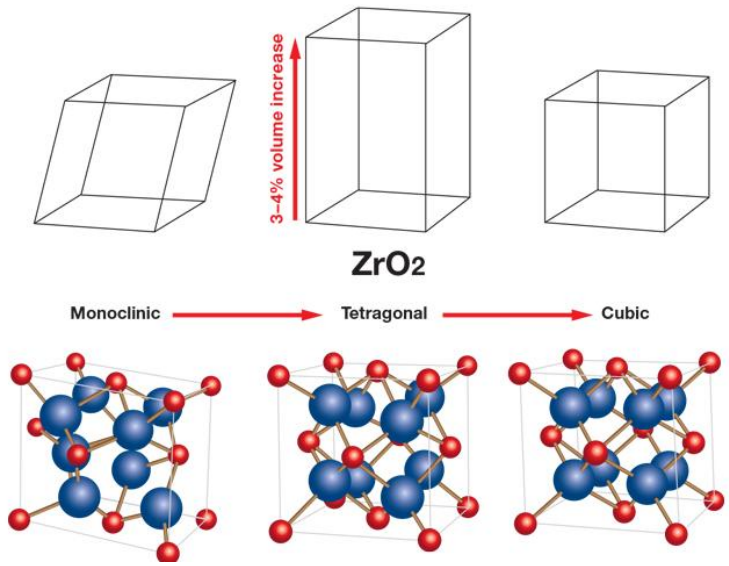
melt $\xrightarrow{2680\ °C}$ c-ZrO_2 $\xrightarrow{2370\ °C}$ t-ZrO_2 $\xrightarrow{1170\ °C}$ m-ZrO_2

problem:

t→m phase transformation leads to a volume exansion of $\Delta V/V_o$=3 %

crystallographic phases of ZrO_2

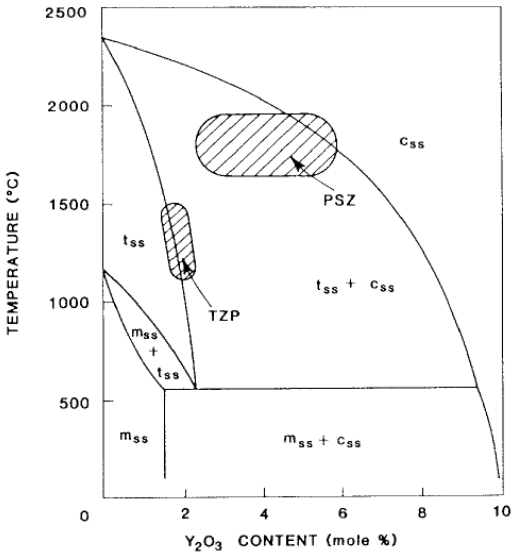
crystal system	space group	space data [Å]	density
cubic a=b=c α=β=γ=90°	F_{m3m}	a = 5.124	6.090 (calculated)
tetrahedral a=b≠c α=β=γ=90°	$P4_2/nmc$	a = 5.094 c = 5.177	6.100 (calculated)
monoclinic a≠b≠c≠a α=γ=90° β>90°	$P2_1/c$	a = 5.156 b = 5.191 c = 5.304 β = 98.9°	5.830



…..how to cope with this?

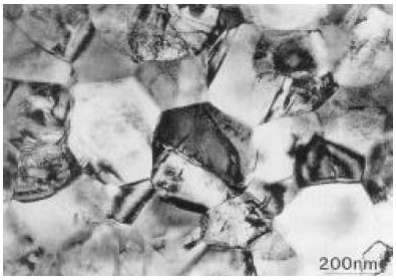
→stabilization of the tetragonal structure!

→partial substitution of Zr^{4+}, e. g. by Mg^{2+}, Ca^{2+}, Ce^{3+}, Y^{3+}



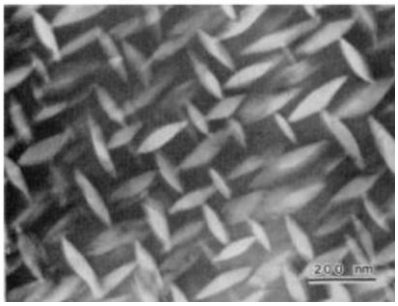
ZrO_2 + 3 % Y_2O_3

TEM image of a Y stabilized TZP



ZrO_2 + <8 % Y_2O_3

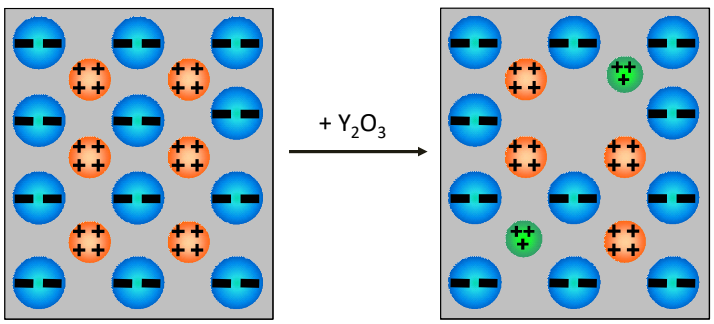
TEM image of a Y stabilized PSZ



TZP	tetragonal zirconia polycrystalline (~ 3 mol % Y_2O_3)
PSZ	partially stabilized zirconia (< 8 mol % Y_2O_3)
FSZ	fully stabilized zirconia (8 – 10 mol % Y_2O_3)

What happens in the crystal structure when Zr^{4+} is partially substituted by Y^{3+}?

doping of ZrO_2 with Y_2O_3



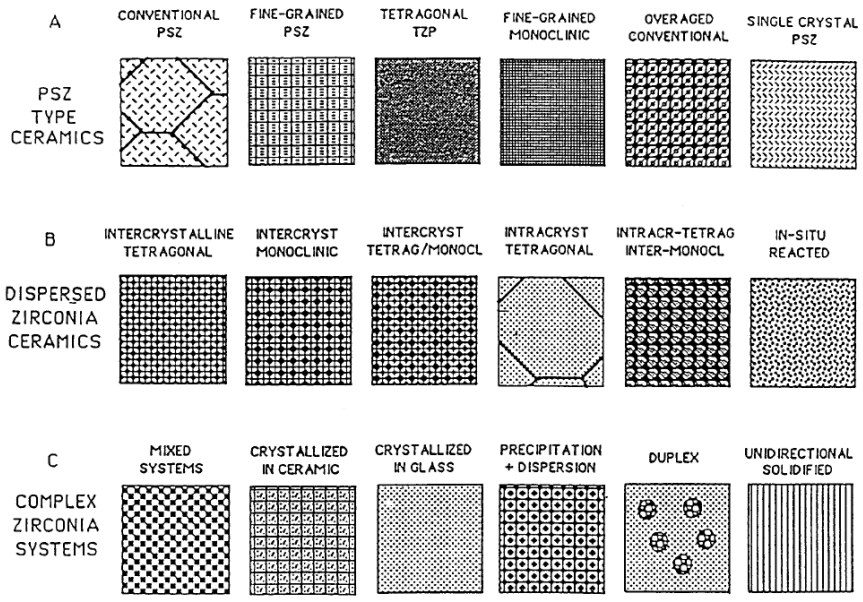
$r_{Zr^{4+}}$ = 59 pm

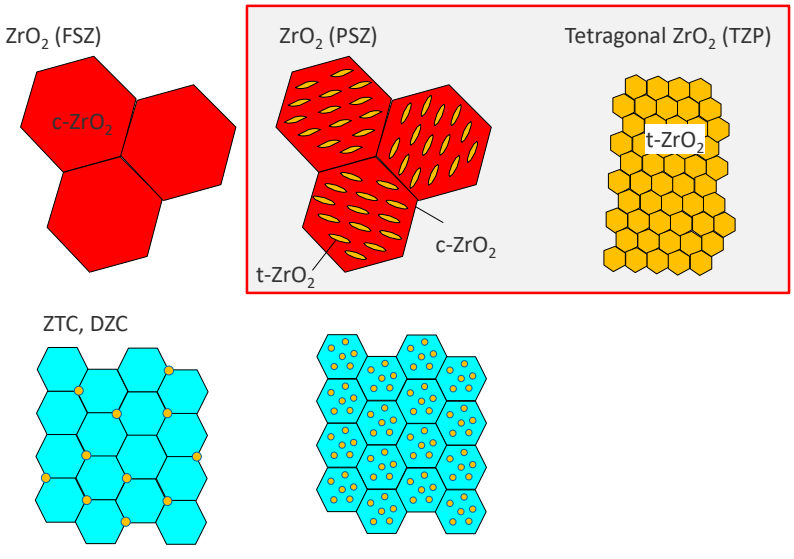
$r_{Y^{3+}}$ = 90 pm

→ massive distortion of the crystal lattice

→ t→m transformation does not occur under ambient conditions

→ defects formed…useable for ion conductivity


A
CONVENTIONAL PSZ
FINE-GRAINED PSZ
TETRAGONAL TZP
FINE-GRAINED MONOCLINIC
OVERAGED CONVENTIONAL
SINGLE CRYSTAL PSZ
PSZ TYPE CERAMICS
B
INTERCRYSTALLINE TETRAGONAL
INTERCRYST MONOCLINIC
INTERCRYST TETRAG/MONOCL
INTRACRYST TETRAGONAL
INTRACR-TETRAG INTER-MONOCL
IN-SITU REACTED
DISPERSED ZIRCONIA CERAMICS
C
MIXED SYSTEMS
CRYSTALLIZED IN CERAMIC
CRYSTALLIZED IN GLASS
PRECIPITATION + DISPERSION
DUPLEX
UNIDIRECTIONAL SOLIDIFIED
COMPLEX ZIRCONIA SYSTEMS


ZrO2 (FSZ)
c-ZrO2
ZrO2 (PSZ)
c-ZrO2
t-ZrO2
Tetragonal ZrO2 (TZP)
t-ZrO2
ZTC, DZC

zirconium dioxide, ZrO_2
(single phase ceramic)

orthopedic:

- hip heads
- bone screws and plates
- knees

dental:

- crowns and bridges
- root sticks
- implant heads

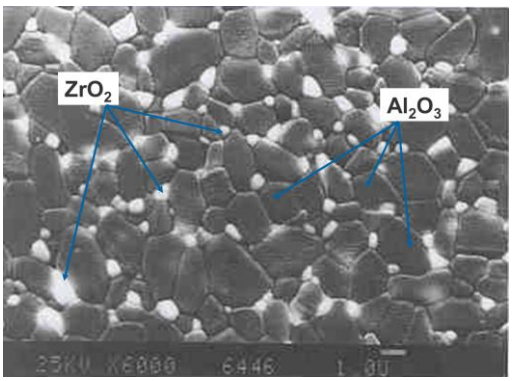


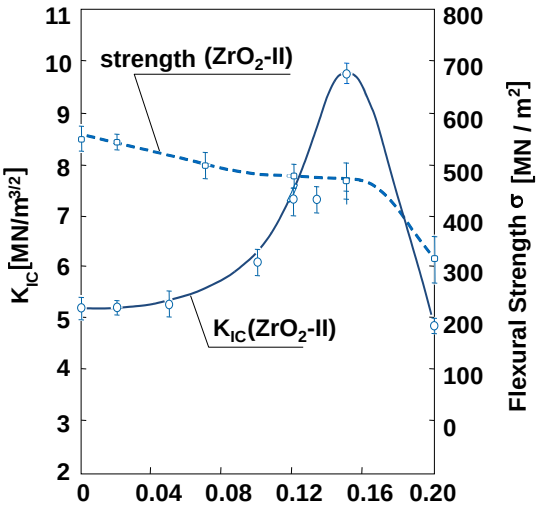






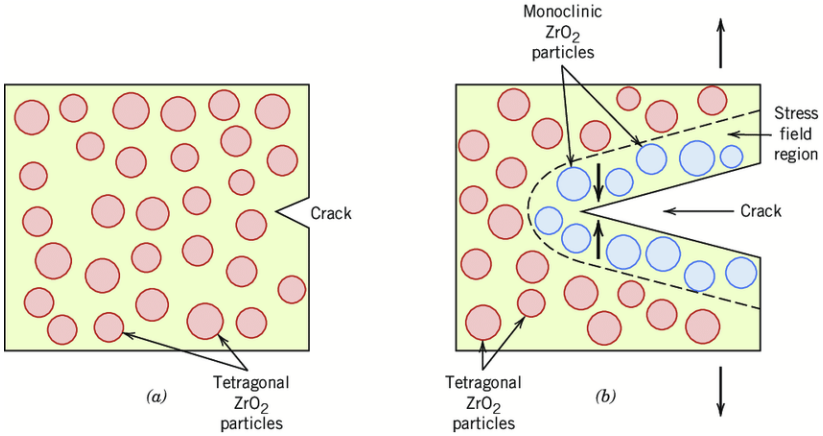
zirconia reinforced alumina
(ZrO_2 as a reinforcement phase for the increase of the fracture toughness)





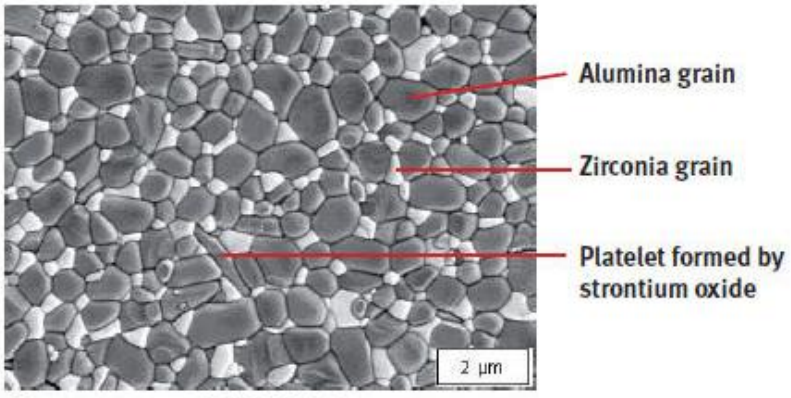
transformation toughening with ZrO_2

How does it work?

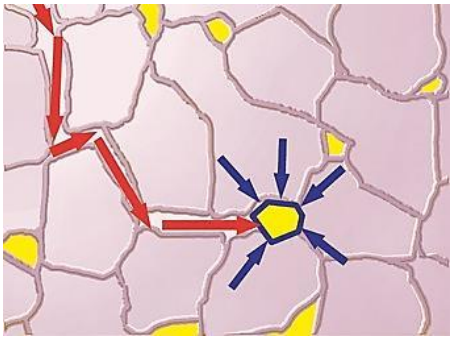


transformation toughening with ZrO_2 combined with platelet reinforcement

How does it work?

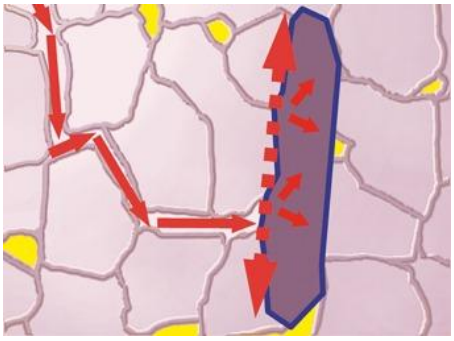


Microstructure of *BIOLOX delta* Material (courtesy of CeramTec AG)



principle of transformation toughening by Y doped ZrO_2 within an Al_2O_3 matrix

dark-pink: Al_2O_3
yellow: ZrO_2



principle of platelet toughening in an alumina matrix

dark: platelet-like crystal

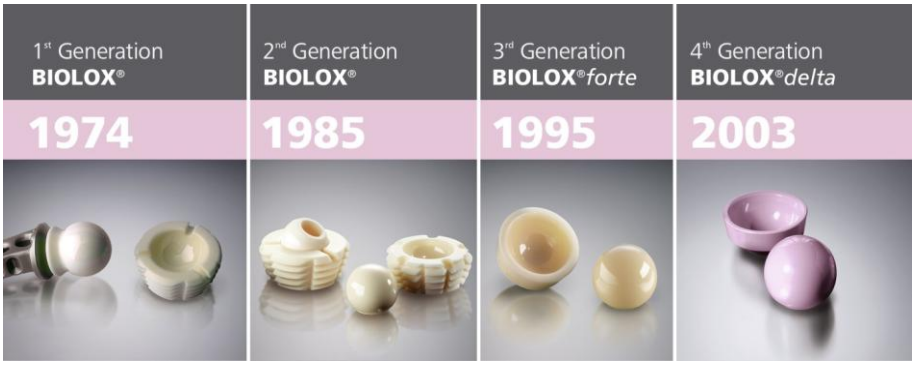
BIOLOX from CeramTec AG, Germany

BIOLOX®*delta* metal free knee implants

- excellent biologigical behavior
- allergy risk not known
- no metal ion release
- no patologic reactions related to ceramic particles
- reduced infection risk
- low amount of polyethylene debries
- high hardness and scatch resistance

all information from manufacturers data sheets

Joint balls and joint socket for hip joint replacement



https://www.ceramtec.de/biolox/duo/









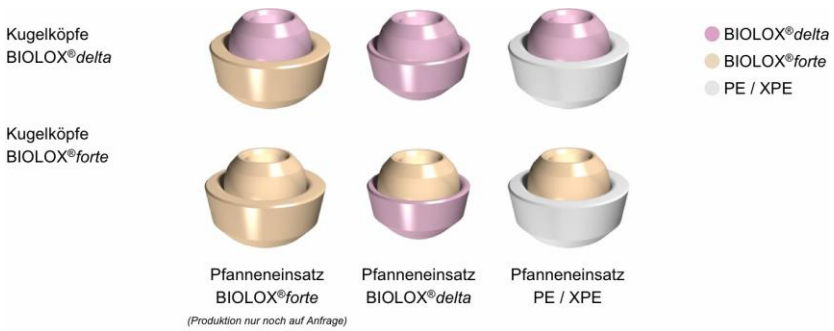
https://www.ceramtec.de/biolox/kniegelenkskomponenten/







shoulder joint implants





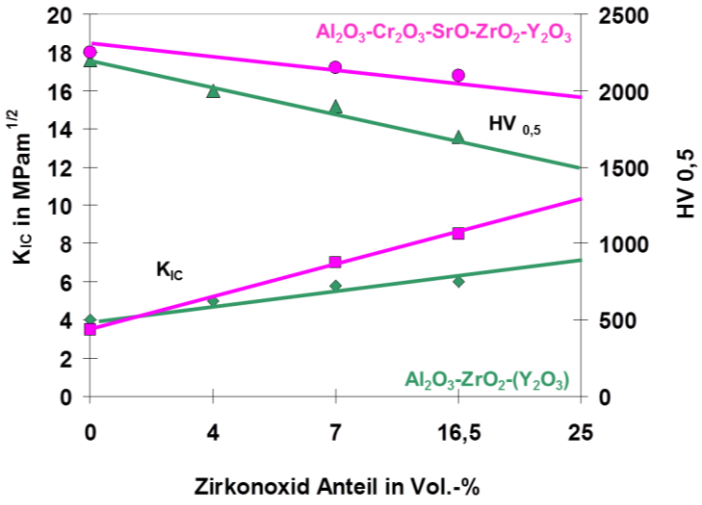
Indication

BIOLOX®DUO bipolar head in case of femur head fracture and head necroses/mortification, when the acetabulum is fairly intact

BIOLOX®DUO is a full-ceramic bipolar system consisting of outer BIOLOX®*forte* sliding bowl (42 mm… 56 mm in 1 mm steps in diametre), BIOLOX®*forte* spherical head ø 28 mm or BIOLOX®*delta* spherical head ø 28 mm and lock ring

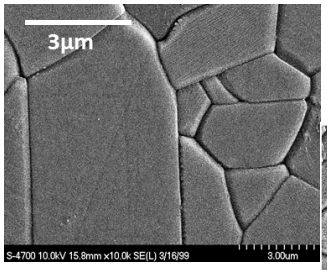
Biolox Delta - composition

82% Al_2O_3
17% ZrO_2
Cr_2O_3, SrO

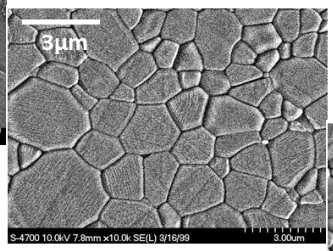


two reinforcement mechanisms for fracture toughness increase

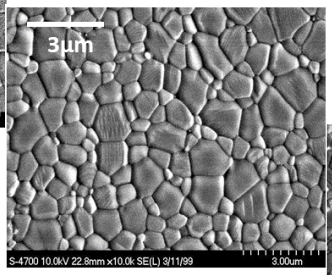
alumina: a microstructure´s brief history



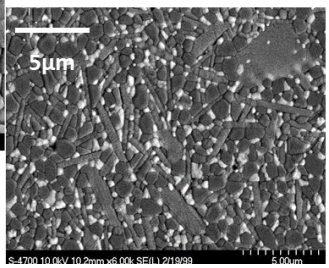
Al_2O_3 from fine-particulate powder
(before 1980)



alumina powders from
the alcoxide process (since 1994)



powders from $NH_4AlCO_3(OH)_2$
decomposition (since 1999)



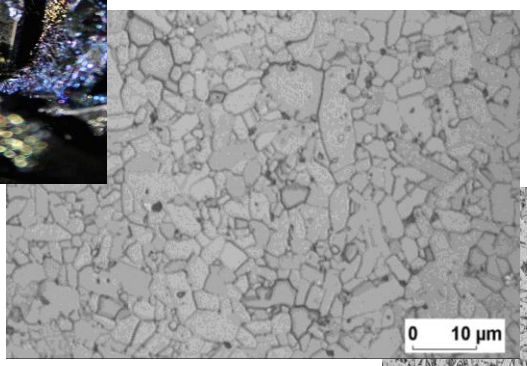
ZrO_2 and platelett
reinforced alumina, ZPTA
(since 1997)

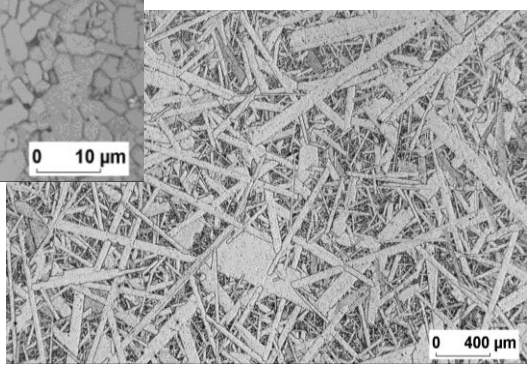
CeramTec

selected Non oxide ceramics → **carbides, nitrides,** borides, …

SiC	burner nozzles, kiln furniture, heating elements
B_4C	superhard abrasives
WC	cutting tools
Si_3N_4	roller bearings, cutting tools, melting crucibles
SiAlONs	cutting tools, thermowells for thermocuppels
AlN	electronic components
BN	superhard abrasives
TiN	wear and erosion resistant coatings
TiB_2	rocket nozzles
ZrB_2	electrodes for metal
$MoSi_2$	heating elements



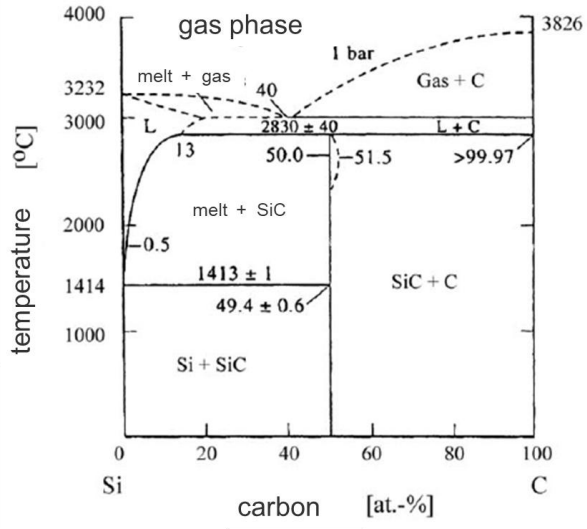




- in 1891 Edward G. Acheson produced a small amount of Silicon Carbide while conducting experiments with the aim of obtaining a hard material from the reaction of clay and carbon.
- he passed a strong electric current from a carbon electrode through a mixture of clay and coke contained in an iron bowl, which served as the second electrode.
- Acheson recognized the abrasive value of the crystals obtained, had them analyzed, found the formula to be SiC, incorporated The Carborundum Company in September 1891, and filed application for a patent on May 10, 1892.
- silicon carbide is also called carborundum, including black and green SiC, both with a shape of hexagonal crystal; black SiC is classified into coke-made and coal-made black SiC, depending on different raw materials.
- SiC is extremely hard and sharp-edged;



	SiO_2	Si_3N_4	SiC
Δ electronegativity	1.54	1.14	0.65
covalent bonding share	68	75	85
formation enthalpy [kcal/mol]	-217	-178	-15



phase diagram of the system Si–C at ambient pressure according to Kleykamp and Schumacher

zirconia **(also: zirkoniumoxid, ZrO_2)**

ρ $\left[\frac{g}{cm^3}\right]$	E [GPa]	K_{IC} $[MPa\sqrt{m}]$	σ_B [MPa]	m [1]
5.89	200	6-10	60-1000	15-25

Härte [HV10]	α $[10^{-6}\ K^{-1}]$	λ $\left[\frac{W}{m \cdot K}\right]$	Schmelzp. [°C]
1300	10	2	2680

crystallographic structures:

melt $\xrightarrow{2680\ °C}$ c-ZrO_2 $\xrightarrow{2370\ °C}$ t-ZrO_2 $\xrightarrow{1170\ °C}$ m-ZrO_2

problem:

t→m phase transformation leads to a volume exansion of $\Delta V/V_o$=3 %

silicon carbide

ρ $\left[\frac{g}{cm^3}\right]$	E [GPa]	K_{IC} $[MPa\sqrt{m}]$	σ_B [MPa]	m [1]
3.2	370	3.5	390	13

hardness [HV10]	α $[10^{-6}\ K^{-1}]$	λ $\left[\frac{W}{m \cdot K}\right]$	decomp. T [°C]
9500	4.3	100	2300

SiC does not melt at high temperatures; instead, it shows decomposition to form the elements Si and C;

decomposition occurs at 2,700 °C (data from 1986)/at 3,070 °C (data from 1998)

Si → Si + C

more properties of SiC

- density: 3.21 g/cm³
- hardness: 9.5
- high abrasion resistance
- relative high bending strength: 390 MPa
- high thermal conductivity: 70-140 W/mK
- low thermal expansion: 4.5 10^{-6} K^{-1} (20 °C…1000 °C)
- high oxidation resistance due to passivation effects: SiO_2-layer on surface
- high corrosion resistance against most acids and bases (corrosion under influence of molten alkalis)
- variation of electric conductivity over wide range: 10^{-2}-10^{13} Ωm
- semiconductor properties by element doping

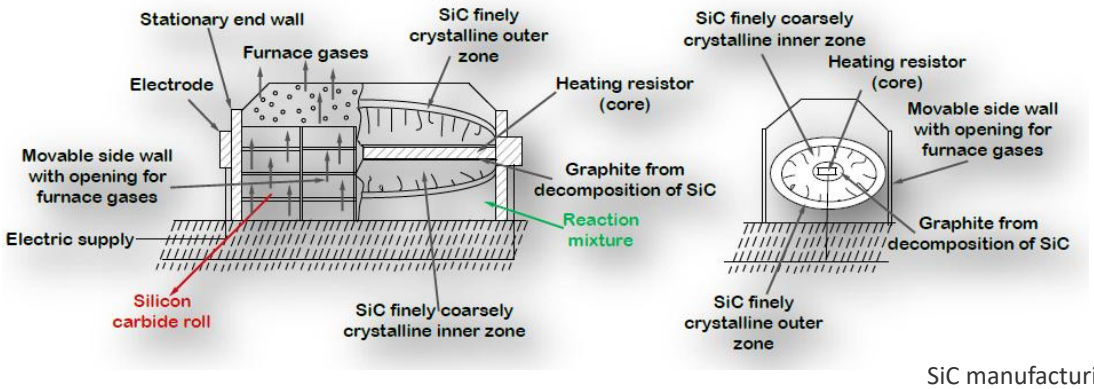
The various reactions involved in the manufacture of silicon carbide are

$SiO_2 + 2C$ (amorphous) $\longrightarrow$ $Si + 2CO$

$Si + C$ (amorphous) $\longrightarrow$ SiC

$SiO_2 + 3C$ (amorph) $\longrightarrow$ $SiC + 2CO$

The temperature was maintained at 2000^{0}C. Higher temperature was prevented as Sic was decomposed into graphite at high temperature.



SiC manufacturing by the Acheson process

The various reactions involved in the manufacture of silicon carbide are

$SiO_2 + 2C$ (amorphous) $\longrightarrow$ $Si + 2CO$

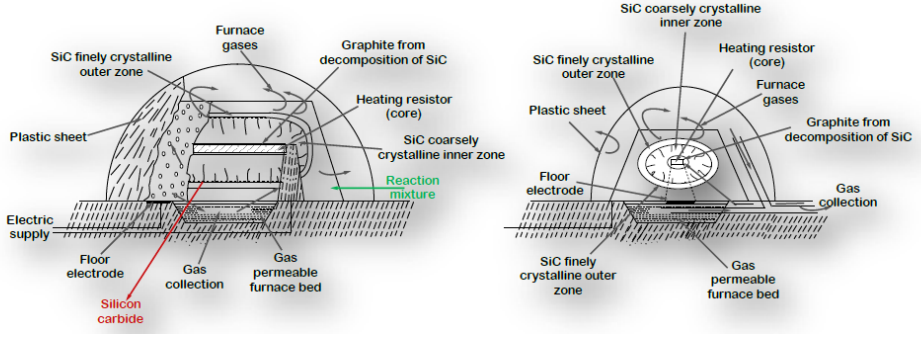
$Si + C$ (amorphous) $\longrightarrow$ SiC

$SiO_2 + 3C$ (amorph) $\longrightarrow$ $SiC + 2CO$

The temperature was maintained at 2000^{0}C. Higher temperature was prevented as Sic was decomposed into graphite at high temperature.

The mixture of 53.5% SiO_2, 40% coke, 5% saw dust and 1.5% salt was charged to furnace until the furnace is fully charged. Saw dusts were added because it increases the porosity of the charge to permit the circulation of vapours. A typical initial current was passed between the two electrodes which were initially 6000amp at 250volts, but its resistance gradually decreased as the reaction proceeds and the final current becomes 20000amp at 75volts. After about 2hr, the current increased rapidly from 6000 to 20000amps and remains steady during the whole run of 36hr. The temperatures at the core are 2200^{0}C and remain steady for the whole 36hr until the reaction was completed. During the reaction CO was generated which was burned at the top of the furnace and escape through a movable side walls.

The ESK furnace consists of floor electrode, gas-permeable furnace bed, gas collector, plastic sheet and heating core. Heating core used for this ESK furnace was horizontal as same in case of Acheson furnace. The mixture of coke and sand was charged by wheel loaders into reaction zone, until the zone was full of raw material or charge reached the top of the graphite columns. Graphite was placed on each electrode of furnace. ESK furnace is considerably larger than Acheson furnace.





Appearance:
When removed from the furnace, the Silicon Carbide is a mass of interlocking iridescent crystals, the crystals themselves being largely twinned and often coalescent. Their iridescence is due to a thin surface layer of silica resulting from superficial oxidation of the carbide. Washing in hydrochloric acid will remove this layer. The crystals vary in colour from very pale green to black depending on the amount of included impurities.

Hardness:
Silicon Carbide was the first material entering the range of hardness between corundum and diamond. It is given the relative position of 9.5 on Mohs scale and diamond at 10.

Properties:
SiC is quite stable chemically. It is stable to acids, not reacting with fuming nitric acid, nor with boiling sulphuric hydrochloric or hydrofluoric acid. Sodium silicate attacks it above 1300ºC, calcium and magnesium oxides attack above 1000ºC. Copper oxide reacts at 800ºC to form the metal silicide. It oxidizes slowly in air above 1550ºC.

Silicon Carbide dissociates in molten iron and the silicon reacts with the metal oxides in the melt. This reaction is of use in the metallurgy of iron and steel.

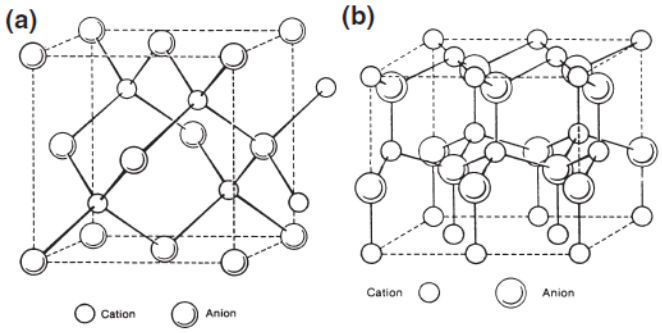
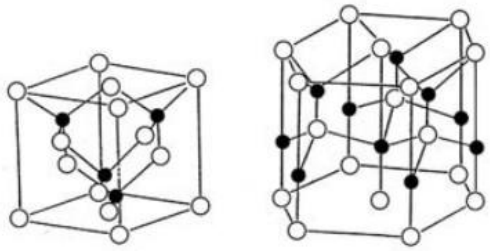


Figure 1.2 SiC Crystal Structures: (a) Zinc Blend Structure for β-SiC and (b) Wurtzite Structure for 6H α-SiC. (From: W.D. Kingery et al, "Introduction to Ceramics", 2nd ed., Wiley, New York, 1976, p. 63).



3C cubic
a=b=c=4.3595 Å
α=β=γ=90°

6H hexagonal
a=b=3.0825 Å
c=15.117 Å
α=β=90°
γ=120°

Edited by A. H. Rashed

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A common system of nomenclature used to describe the different crystalline polytypes assigns a number corresponding to the number of layers in the unit cell followed by a letter suffix designating the crystal symmetry; "C" for cubic, "H" for hexagonal and "R" for rhombohedral. The most common SiC polytypes are the 3C, 4H, 6H, 15R and 9T. The cubic 3C is commonly referred to as beta silicon carbide, β-SiC, which has the zinc blend structure, while all other polytypes are referred to as alpha silicon carbide, α-SiC. In general, α-SiC phase is mainly 6H, which is a wurtzite structure. Figure 1.2 shows an illustration of the two major crystal structures, zinc blend and wurtzite, exhibited by the two SiC phases. Phase transformation of SiC occurs from β-SiC to α-SiC upon heating. Undoped β-SiC transforms to 6H and 15R above 1650°C with the 15R being a metastable phase that transforms to 6H. Doping β-SiC with boron lowers the transformation temperature and results in the formation of 4H polytype. On the other hand, doping with nitrogen prevents the formation of 4H and stabilizes the 6H. The $\beta \rightarrow \alpha$ phase transformation is irreversible under ambient atmosphere. However, under pure nitrogen atmosphere, the transformation can be reversed and β-SiC phase can be stabilized up to 2500°C by applying a nitrogen pressure.

Edited by A. H. Rashed

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Under atmospheric pressure, silicon carbide does not melt when heated to elevated temperatures rather, it sublimes and/or dissociates. In addition, incongruent melting of SiC was reported at 2829°C under >500 psi pressure of argon and is also possible when it is heated rapidly in an arc-image furnace at atmospheric pressure.

- Synthesis of raw SiC

  → Future process technologies:

  - General electric process:

    Generation of a silica gel in a sugar solution
    Dehydration at 300 °C
    Thermal treatment at 1800 °C in Ar → β-SiC
  - Pyrolysis of rice hulls

    Contain C (cellulose) and amorphous silica
    Thermal treatment between 1290 °C and 1600 °C
  - Fluidized bed technology

    Continuous process for production of β-SiC
    1500-1600 °C
  - Rotary furnace

    Graphite tube, 1600-1700 °C

→ Other syntheses techniques/manufacturing technlogies:

- Elemental synthesis from Si and C
- Reaction of Si with methane
- Reaction of silanes with hydrocarbons
- Reaction of $SiCl_4$ with hydrocarbons
- Reaction of metalorganic precursors

• Pyrolisis of silanes

$$CH_3SiCl_3 \rightarrow \beta\text{–}SiC + 3HCl$$

• Gas phase reactions

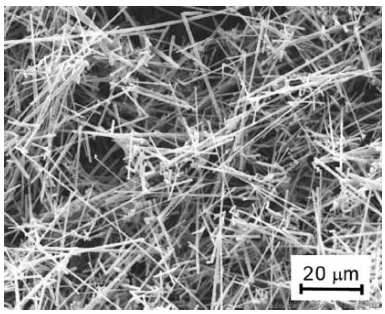
$$SiCl_4 + CH_4 \rightarrow \beta\text{–}SiC + 4HCl$$

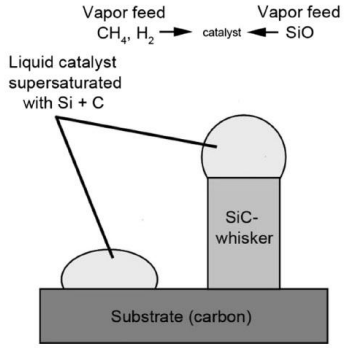
• 10 times finer powders but extremely high price

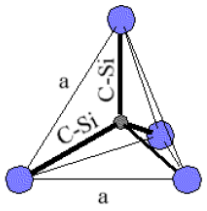
- Whiskers, platelets and fibres of SiC

  → Whisker

  - Vapour solid reaction (VS)
  - Evaporation/vapour condensation (EV)
  - Vapour-liquid-solid reaction (VLS)







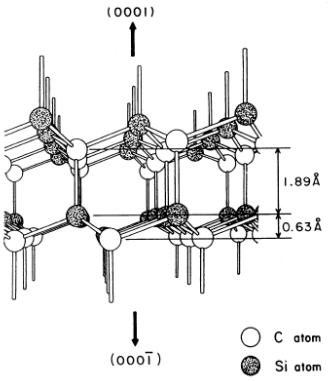
Si atom

C atom

Fig. 4. The tetragonal bonding of a carbon atom with the four nearest silicon neighbours. The distances a and C-Si are approximately 3.08Å and 1.89Å respectively.

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Silicon face



Carbon face

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Silicon carbide is made up of equal parts silicon and carbon.

Both are period IV elements, so they will prefer a covalent bonding such as in the left figure. Also, each carbon atom is surrounded by four silicon atoms, and vice versa.

This will lead to a highly ordered configuration, a single crystal, such as in the right figure.

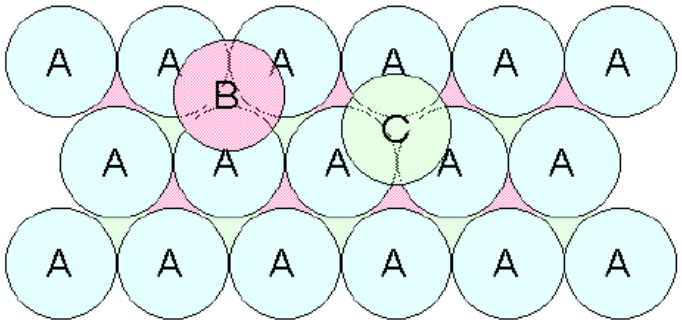
Since there is exactly one silicon atom for each carbon atom, lets think of this as a unit, symbolized by this ball.

The most efficient way to pack balls is in this hexagonal fashion, in crystallographical terms called Hexagonal Close Packing.

This layer of balls labeled A represent a double layer of silicon and carbon atoms.

The next layer has to be displaced according to the covalent bonding scheme, but we find there are two choices, indicated by B and C respectively.

Carl-Mikael Zetterling



The rule is A can be followed by B or C, but not A. The same goes for B and C.

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It turns out that there exists several stable stacking orders with a long term order throughout a large sample.

This figure shows four of them:
3C, 2H, 4H and 6H. The number corresponds to the number of double layers of Si and C before the pattern is repeated. For instance, 4H repeats ABAC ABAC etc.

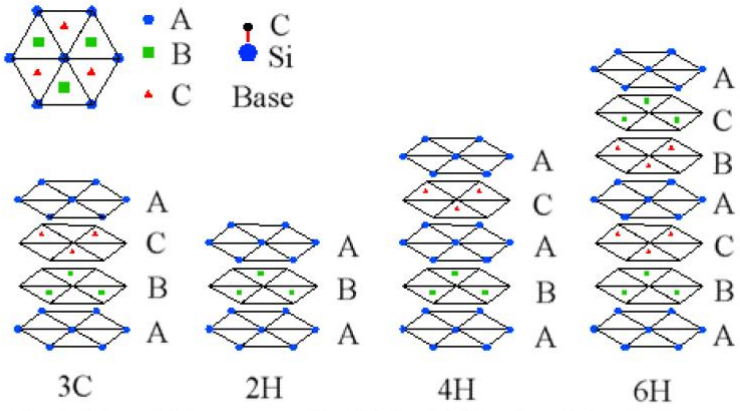
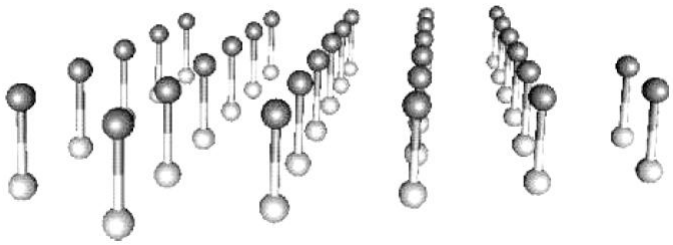


Fig. 5. The stacking sequence of double layers of the three most common SiC polytypes.

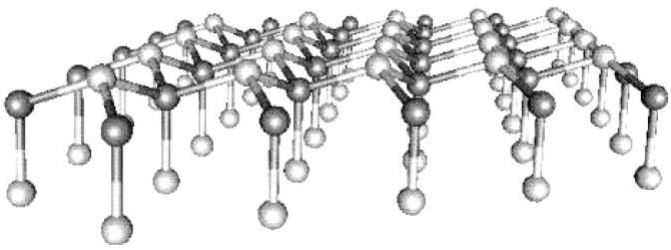
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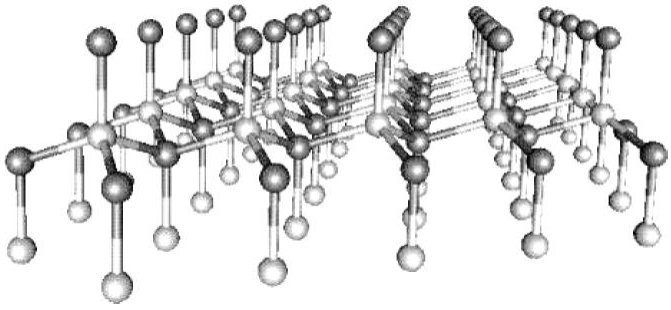
a)

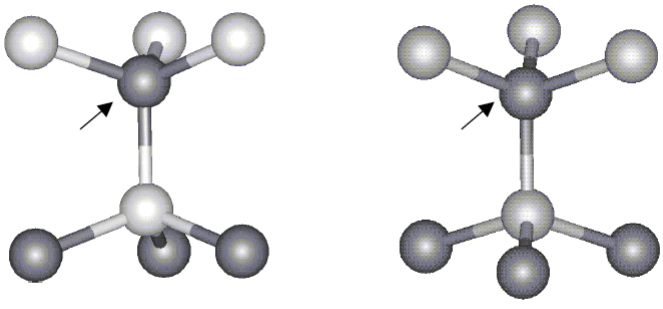


b)



c)

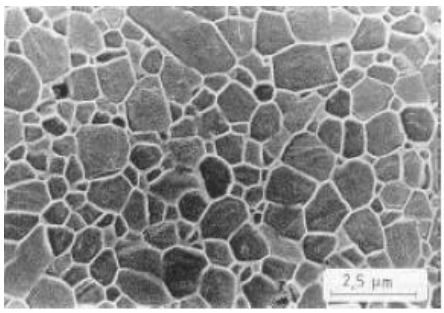




carbon atom in hexagonal position (left), carbon in cubic position right

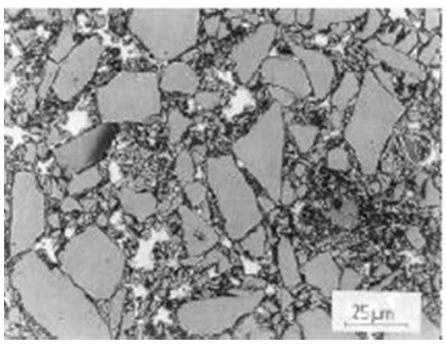
types of SiC materials received by different sintering methods

- sintered SiC – SSiC
- liquid phase sintered SiC – LPSSiC
- hot pressed SiC – HPSiC
- hot isostatically pressed SiC – HIPSiC
- ceramically bonded SiC – CBSiC
- nitride-bonded SiC – NBSiC
- re-crystallized SiC – RSiC
- reaction-bonded SiC – RBSiC
- silicon infiltrated SiC – SiSiC



SSiC:
dense due to glassy phase dank

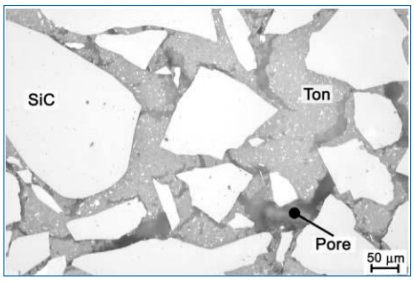
liquid-phase sintered SiC
grain boundaries: white

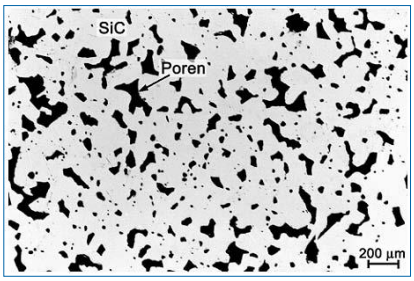


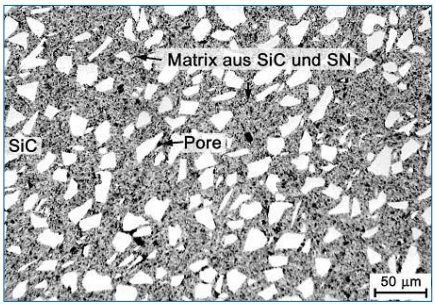
SiSiC:
dense due to liquid-metal infiltration

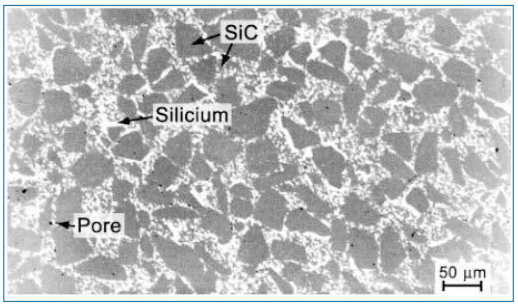
SiSiC; metal infiltrated SiC;
pore-filling metal appears bright

reduced temperature resistance:
$F_{p,\ Si}$=1410 °C







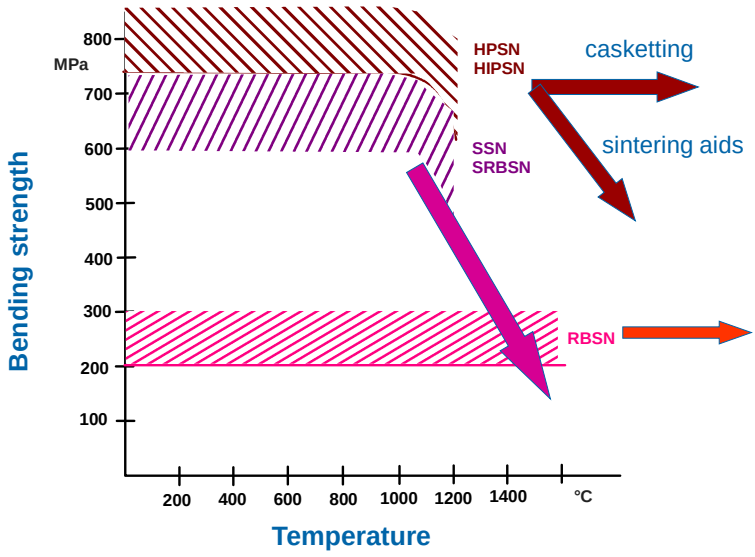


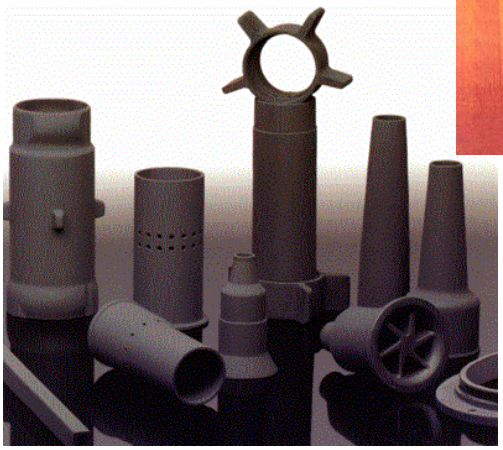
properties of different SiC types

	RSiC	SiSiC	SSiC	HPSiC	HIPSiC
Dichte [g/cm^3]	2.5-2.9	3.10	3.15	3.20	3.21
Biegefestigkeit [MPa]	80-100	400	500	600-700	600-700
E-Modul [GPa]	240	370	390	420	450
K_{IC} [MPa$\sqrt{m}$]	-	3-4	4-5	5-6	5-6
Weibullmodul [1]	-	10	10	10	10-15
Wärmeleitfähigkeit $\left[\frac{W}{m \cdot K}\right]$	-	120	70	90	90-120
Wärmedehnung [10^{-6} K^{-1}]	4.5	4.4	4.5	4.6	4.5
Elektrischer Widerstand $\left[\frac{\Omega}{cm}\right]$	10^{11}	10	10^3	10^5	10^5
Offene Porosität [%]	25	0	1	0	0
Obere Anwendungstemp. [°C]	2000	1400	1700	1300	1300

applications

- abrasive,
- grinding media
- cutting tools
- heat exchangers
- heating elements
- HT furnaces internals
- light emitting diodes
- thyristors and circuit breakers
- catalyst supports

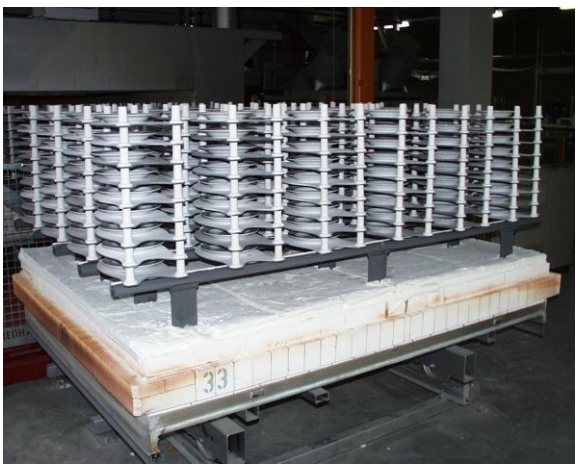


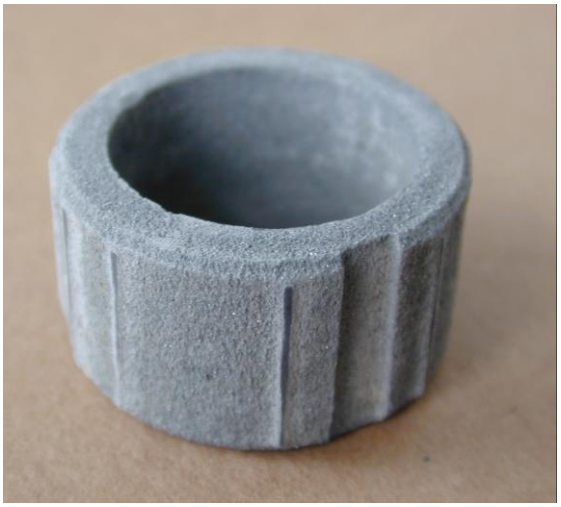


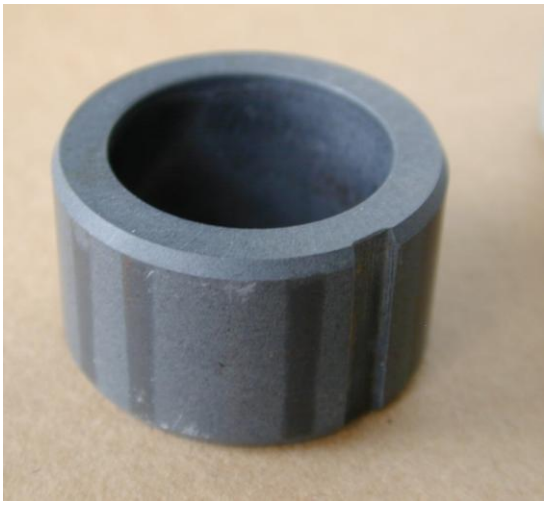





CarSIK-NG 08/01


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Silicon nitride, Si_3N_4

$HPSi_3N_4$

ρ $\left[\frac{g}{cm^3}\right]$	E [GPa]	K_{IC} $[MPa\sqrt{m}]$	σ_B [MPa]	m [1]
3.2	300	7	900	15

Härte [HV10]	α $[10^{-6}\ K^{-1}]$	λ $\left[\frac{W}{m \cdot K}\right]$	Zersetzung [°C]
1500	3.1	25	1900

manufacturing processes*

- SN
- SSN
- GPSSN
- HPSN
- HIPSN
- RBSN

* see also SiC

	RBSN	SSN	HPSN
Rel. Dichte [%]	65-85	95-100	98-100
Biegefestigkeit bei RT [MPa]	200-350	700-1400	700-1500
K_{IC} $[MPa\sqrt{m}]$	1.5-3	5-10	5-10
Härte [GPa]	5-10	14-18	14-18
Wärmeleitfähigkeit $\left[\frac{W}{m \cdot K}\right]$	4-15	20-40	20-40
Wärmedehnung $[10^{-6}\ K^{-1}]$	2.5-3.0	2.5-3.8	2.5-3.8

Si_3N_4 properties

- extremely high strength
- high fracture toughness
- excellent wear resistance
- high thermal conductivity
- very low thermal expansion
- thermal shock resistance



applications

- internal combustion engines valves
- ball bearings
- mechanically loaded parts in HT applications
- coating on AFM-cantilever tips



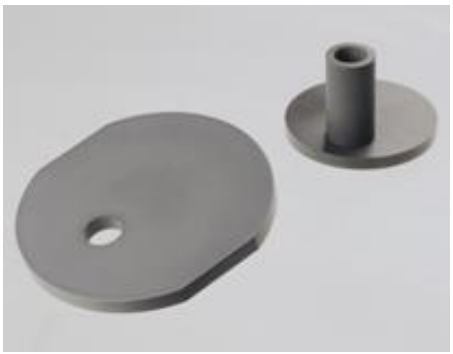


H.C. Starck Ceramics

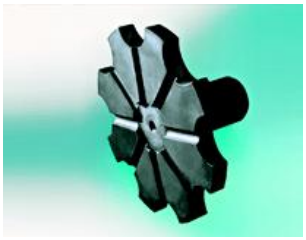
properties

- high thermal shock and corrosion resistance
- low wettability with non-ferrous metal castings
- high thermal insulation capabilities

dosing and riser pipes



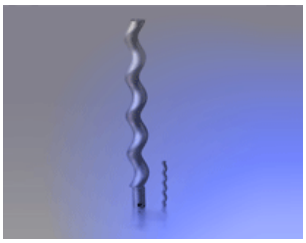
nozzles and valves for flow control



impeller for cleaning non-ferrous metal melts



extrusion dies for cold forging and warm forming



rotors for progressive cavity pumps

more examples: http://www.hcstarck-ceramics.de